

FitBuddy: Frontend Development Guide

AI Fitness Plan Generator using Gemini Models | Epic 4 Implementation Document

1. Overview of Frontend Development (Epic 4)

Epic 4 focuses on designing and developing a user-friendly, responsive, and visually structured interface for *FitBuddy*. Using HTML, CSS, and Jinja2 templating, the frontend allows users to input their fitness parameters, submit them to the FastAPI backend, and view customized AI-generated workouts and nutrition plans dynamically.

Key Milestone Objective: Build clean HTML forms and interactive layout templates rendered through Jinja2, ensuring an intuitive user experience and clean integration with backend route responses.

2. Core Components of Frontend Structure

- **Layout Templates (*base.html*):** A master layout containing common header, footer, stylesheet links, and responsive viewport meta tags.
- **User Input Form (*index.html*):** An interactive form capturing age, weight, height, fitness goals, dietary preferences, and activity level.
- **Results Dashboard (*plan.html*):** A structured view displaying the Gemini-generated fitness regimen and workout schedules.
- **Static Stylesheet (*style.css*):** Custom CSS for modern styling, form controls, responsive grids, and clean typography.

3. Reference Code: *index.html* (User Input Form)

Below is the standard HTML template for collecting user fitness details:

```

<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>FitBuddy - AI Fitness Plan Generator</title>
  <link rel="stylesheet" href="{{ url_for('static', path='/style.css') }}">
</head>
<body>
  <div class="container">
    <header>
      <h1>FitBuddy <span>AI Fitness Generator</span></h1>
      <p>Powered by Google Gemini Models</p>
    </header>
    <form action="/generate" method="POST">
      <label for="age">Age:</label>
      <input type="number" id="age" name="age" required>

      <label for="goal">Fitness Goal:</label>
      <select id="goal" name="goal">
        <option value="Weight Loss">Weight Loss</option>
        <option value="Muscle Gain">Muscle Gain</option>
        <option value="Endurance">Endurance & Stamina</option>
      </select>

      <button type="submit">Generate Plan</button>
    </form>
  </div>
</body>
</html>

```

4. Reference Code: style.css

Clean and modern styling ensures the interface remains intuitive and fully responsive across devices:

```

body {
  font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif;
  background-color: #f0f2f5;
  color: #333;
  margin: 0;
  padding: 20px;
}
.container {
  max-width: 600px;
  background: #ffffff;
  margin: 40px auto;
  padding: 30px;
  border-radius: 8px;
  box-shadow: 0 4px 12px rgba(0,0,0,0.1);
}
header h1 {
  color: #11998e;
  margin-bottom: 5px;
}
form label {
  display: block;
  margin-top: 15px;
  font-weight: 600;
}
form input, form select, form button {
  width: 100%;
  padding: 10px;
  margin-top: 5px;
  border: 1px solid #ccc;
  border-radius: 4px;
  font-size: 1rem;
}
form button {
  background: linear-gradient(135deg, #11998e, #38ef7d);
  color: white;
  border: none;
  margin-top: 25px;
  cursor: pointer;
  font-weight: bold;
}
form button:hover {
  opacity: 0.9;
}

```

5. Summary of Frontend Delivery

With Jinja2 templating integrated into FastAPI (*app.py* and *routes.py*), user inputs are seamlessly transmitted to the Gemini AI backend, instantly returning a dynamic and styled fitness roadmap.